

Sathwick Reddy Yalla

Senior Software Engineer

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 sathwickreddyy.github.io/sathwick-portfolio

PROFESSIONAL SUMMARY

Senior Software Engineer with **5+ years of experience** building scalable backend platforms, distributed workflows, cloud-native services, and data-intensive systems at Morgan Stanley, Amazon, and Oracle. Strong in Python, FastAPI, Java, Spring Boot, AWS, REST APIs, event-driven architecture, and modular backend design, with hands-on experience across pandas, Polars, and production data pipelines. Proven record of designing reusable frameworks, reducing delivery effort by up to 95%, driving ~\$1.2M annual cost savings, and improving engineering velocity. Experienced in AI-assisted engineering using LLM agents, spec-driven development, multi-agent workflows, and Git worktrees.

PROFESSIONAL EXPERIENCE

Morgan Stanley, Senior Software Engineer

07/2024 – Present
Bengaluru, India

- Designed and implemented a **reusable processing and publishing framework** for onboarding data-intensive business modules, reducing module delivery time by **~72%**.
- Delivered a forecasting and scenario-analysis platform in 2 weeks versus a planned 2-month timeline by applying **spec-driven development**, reusable backend components, and AI-assisted execution.
- Built scalable backend services** and workflow components using Python, FastAPI, pandas, Polars, Java, Spring Boot, and AWS to support analytical processing and business-critical decision workflows.
- Established **AI-assisted engineering workflows** using Claude, Codex, Copilot, Cursor, Amp, subagents, multi-agent execution, and parallel Git worktrees to accelerate implementation, refactoring, and documentation.
- Performed deep system analysis across forecasting services, processing engines, publishing workflows, and downstream integrations, improving maintainability, delivery confidence, and production readiness.
- Integrated forecasting applications with scenario-based prediction workflows, enabling users to model business outcomes and make faster data-backed decisions.
- Created detailed **LLD/HLD documentation**, technical flow diagrams, onboarding material, and knowledge-sharing assets to improve team execution and long-term maintainability.
- Improved **risk, security, and controls posture** by implementing and documenting SSL-compliant application changes.
- Partnered with cross-functional teams in a fast-paced delivery environment to ship high-priority backend and analytics initiatives.

Amazon, Software Development Engineer / Programmer Analyst I

02/2022 – 06/2024
Bengaluru, India

- Designed and implemented a generic AWS Step Functions construct using cost-effective cloud-native patterns, generating **~\$1.2M in annual cost savings**.
- Built a reusable **distributed workflow orchestration model** across AWS services, reducing new project setup effort by **~95%**.
- Developed a **generic constructs library** with reusable infrastructure components and high-quality coding standards, adopted by **10+** internal clients.
- Designed and implemented an innovative end-to-end scalable **grafted tree paradigm model**, improving project setup efficiency by **~70%**.
- Built and maintained cloud-native distributed systems using AWS Step Functions, S3, SQS, SNS, EventBridge, ECS, EC2, IAM, CloudFormation, AWS CDK, and CI/CD pipelines.
- Improved engineering quality by introducing **reusable infrastructure patterns**, automated setup flows, and end-to-end testing practices.
- Resolved and debugged **40+** high-priority production and platform tickets within a 7-day window.

- Conducted internal training sessions on **AWS technologies**, platform patterns, and reusable architecture practices.

Oracle, Application Developer / Software Engineering Intern

01/2021 – 02/2022
Hyderabad, India

- Worked as a **Java backend developer** in the Business Analytics team, contributing to the RPD Generator Framework Backend and Oracle Analytics Services.
- Developed and maintained backend functionality for analytics metadata generation, visualization enablement, and business reporting workflows.
- Debugged and resolved software defects across **on-premise and Oracle Analytics Cloud** environments.
- Optimized SQL query execution time by **~43%** through **map-side join optimization** and query restructuring.
- Modernized legacy backend components by modularizing older code paths, improving maintainability and long-term extensibility.

CORE SKILLS

Languages

Python, Java, TypeScript

Backend

FastAPI, Spring Boot, REST APIs, Microservices, Async Workflows, Event-Driven Systems, Modular Architecture

Cloud & Infrastructure

AWS, Docker, Jenkins, AWS CDK, Terraform, CI/CD

Databases & Storage

PostgreSQL, DynamoDB, Redis, SQL, S3

AI Engineering

LiteLLM, Vercel AI SDK, Amazon Bedrock, Spec-Driven Development, Claude, Codex, Copilot, Cursor, Amp by Sourcegraph, Git Worktrees, Subagent Workflows, Multi-Agent Development.

Frontend

React, Next.js

EDUCATION

Bachelor of Engineering, Computer Science,
Chaitanya Bharathi Institute of Technology
08/2017 – 05/2021 | Hyderabad, India | CGPA: 8.73

CONTRIBUTIONS

Buzz Women Tech Philanthropy  ,
Opensource Contribution
05/2025 – 11/2025 | Remote

- Designed serverless AWS CDK infrastructure for automated email notifications, cost analysis, and volunteer enablement.
- Presented technical knowledge-sharing sessions on reusable frameworks, financial forecasting systems, backend architecture patterns, and cloud-native delivery practices.